

Water Harvesting: Structures and Practices

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1. What is Water Harvesting?

- The collection and storage of rainwater that runoff from roof tops, parks, roads, and open grounds.
- Objective: To store water for future use and recharge groundwater.
- Concept: Reducing the speed of runoff to increase infiltration into the soil.

2. Rooftop Rainwater Harvesting

- Popular in urban areas where land space is limited.
- Uses the existing roof of a house or building as a catchment area.
- Water is channeled through PVC pipes to a storage tank or recharge pit.

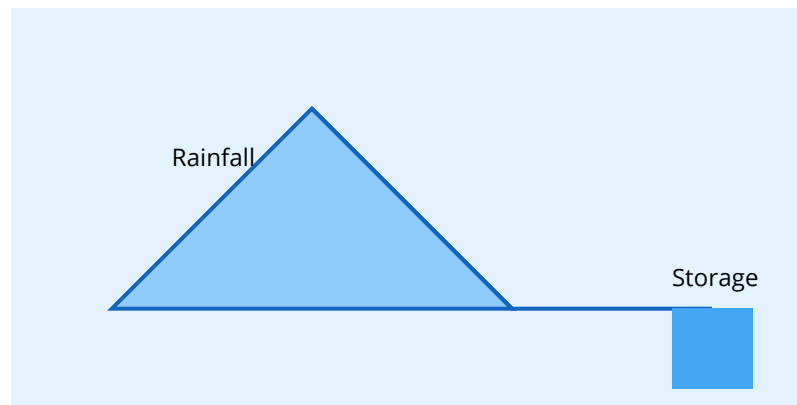


Diagram: Rooftop Rainwater Harvesting Process

3. Traditional: The Tanka System

- Traditional to the Thar Desert region of Rajasthan.
- Underground cylindrical tanks built to store rainwater from rooftops or artificial catchments.
- Keeps water cool and provides drinking water during the hot summer months.

| 4. Traditional: Johads

- Small earthen check dams that capture rainwater runoff.
- Primary purpose: To recharge groundwater and improve the local water table.
- A community-driven approach common in Alwar, Rajasthan.

5. Check Dams and Anicuts

- Small barriers built across a stream to slow down the flow.
- Reduces soil erosion and allows water to seep into the ground.
- Commonly used in hilly terrains and small river channels.

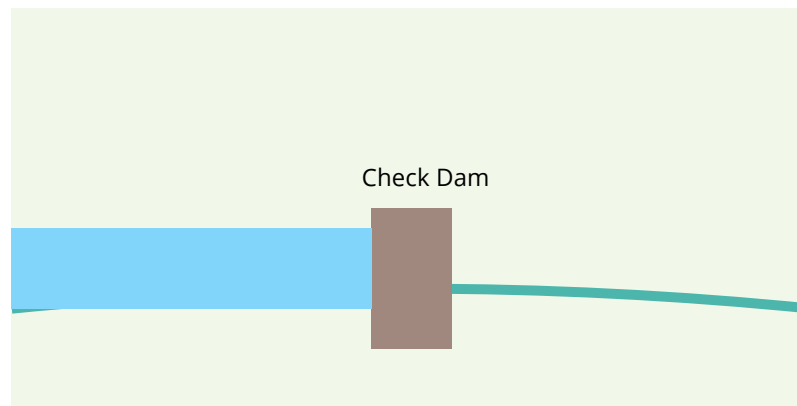


Diagram: A Stone Check Dam Structure

| 6. Percolation Tanks

- Man-made surface water bodies created on permeable land.
- Stores surface runoff to allow it to percolate into the aquifer.
- Helps in improving the yield of nearby bore-wells.

| 7. Gabion Structures

- Wire mesh baskets filled with stones placed across small streams.
- Prevents soil erosion while letting water pass through slowly.
- Cost-effective and requires low maintenance.

| 8. Farm Ponds

- Small dug-out structures in agricultural fields.
- Used for collecting excess runoff during monsoons.
- Provides critical irrigation to crops during dry spells.

| 9. Recharge Shafts and Wells

- Techniques used to guide water directly into deep underground aquifers.
- Effective when the top soil layer is impermeable (clayey).
- Must include a filter layer to prevent contamination.

| 10. Contour Bunding

- Building small embankments along the same elevation level (contour).
- Used on sloping land to stop water from flowing downhill too fast.
- Increases soil moisture and prevents topsoil loss.

| 11. Bamboo Drip Irrigation

- Ancient practice in Meghalaya.
- Uses bamboo pipes to transport water from perennial springs to fields.
- An extremely efficient and low-cost gravity-based system.

| 12. Zabo System

- Practiced by the Chakhesang tribe of Nagaland.
- Includes forest protection, water storage tanks, and livestock management.
- Water flows from the forest into storage ponds then to paddy fields.

| 13. In-situ Water Harvesting

- Capturing rain exactly where it falls in the field.
- Methods include mulching, deep plowing, and leveling.
- Directly improves crop yield by keeping the soil moist.

| 14. Benefits of Water Harvesting

- Reduces the cost of water supply for households.
- Mitigates the effects of droughts.
- Reduces urban flooding by decreasing storm-water runoff.

| 15. Maintenance and Safety

- Cleaning of catchments (roofs) before the first rain.
- Regular cleaning of filters and storage tanks.
- Testing water quality before using it for drinking purposes.

| 16. Government Initiatives

- JalShakti Abhiyan (Catchthe Raincampaign).
- Mandatory rooftop harvesting in states like Tamil Nadu.
- Subsidies provided for farm ponds and recharge wells.

| 17. Conclusion

- Waterharvestingisno longer an option; it is a necessity.
- Combining traditional wisdom with modern technology.
- Community involvement is the key to success.

18. Exam Practice: Question 1

- **Q:** Which state in India was the first to make rooftop rainwater harvesting mandatory for all buildings?
- **Answer:** Tamil Nadu. This policy has significantly helped in reviving the groundwater levels in cities like Chennai.

19. Exam Practice: Question 2

- **Q:**What is the primary function of a 'Johad'?
- **Answer:** The primary function of a Johad is to capture rainwater runoff and allow it to percolate into the ground to recharge local aquifers and wells.

20. Exam Practice: Question 3

- **Q:** Describe the 'Tanka' system found in Rajasthan.
- **Answer:** A Tanka is an underground storage tank, often paved and covered, used to collect rainwater from a 'Paitan' (catchment area). It is a vital source of drinking water in arid regions.